

FR-017 M0 — Agent Evaluation & Observability Engineering Spike

FR017 — `sources/optional/m0_spike.md`

Masterclass source package snapshot

Functional Requirement	FR-017
Edition	Engineering Learning Academy — Source Package snapshot
Status	**Complete (M0 spike)** — **Accepted** ; unlocked M1 under narrow scope;
Audience	Academy generation and study (snapshot; SoT remains authoritative)
Document type	Source package snapshot (PDF)

Faithful rendering of a regenerable Masterclass source snapshot. Do not hand-edit; regenerate via `python scripts/build_masterclass_package.py`. Repository documentation remains the source of truth.

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FR-017 M0 — Agent Evaluation & Observability Engineering Spike

Status: **Complete (M0 spike)** — **Accepted**; unlocked M1 under narrow scope; FR-017 later **Complete / Frozen** — [fr017_agent_evaluation_observability.md](#)

Date: 2026-08-07

Phase: Horizon 1A Stage 11

Preceding: [FR-016 acceptance](#) (Complete / Frozen / Accepted); [ADR-008](#); [FR-015 acceptance](#); [ADR-007](#)

Scope (M0): Document-only architecture spike. **No production code.**

Does not begin (M0): M1 implementation, dashboards, frameworks, Playwright, FR-016 redesign, Horizon 1B, Masterclass generation.

Owner direction accepted for this spike:

1. Preserve the **derive-only** constraint.
2. Strengthen **audit-reconstructability** criteria.
3. **Do not allow FR-017 to block Horizon 1B.**

1. Executive summary

FR-017 as written in the functional specification is a **laundry list** that largely duplicates work already frozen in FR-008, FR-014, FR-015, and FR-016.

Finding: The only genuine remaining gap is a **thin orchestration-layer evaluation substrate** — derived metrics over existing `OrchestrationRun` / `Handoff` / `child AgentRun` audits, plus packaging of offline corpus / fault-injection patterns already proven in FR-016.

Finding: Metrics and reconstructability do **not** require new DOS events, new SoTs, ToolPolicy changes, or dashboards. Inventory (\$4) shows 1:1 mapping from existing fields.

Finding: Horizon 1A **application usability** is already satisfied by FR-008–FR-015 (with FR-016 optional learning proof). Gating Horizon 1B on FR-017 is process fiction and is **rejected**.

M0 recommendation: **GO to M1 under narrow scope** (learning / substrate posture, derive-only, reconstructability-first). **NO-GO** on the full laundry list, dashboards, framework observability, and 1B blocking.

2. Engineering problem (restated)

How do we **evaluate and compare** multi-agent orchestration runs using evidence **already present** in append-only audits, such that:

1. an owner or engineer can reconstruct the full authority story without guessing;
2. corpus regressions stay offline and deterministic;

3. we do not invent a second system of record or an observability product?

This is **evaluation as derived views**, not instrumentation theatre.

3. Dual-value and commercial honesty

Test	Result for narrow FR-017
Improve interview/offer odds?	Indirect at best
Reduce job-search effort?	Marginal (show/history already exist)
Required infrastructure?	Modest — eval harness for multi-agent substrate

Near-term commercial value: low.

Learning / interview-transferable value: high if scoped as anti-theatre evaluation (derive over audits; refuse dashboards).

Daily preparation remains `cic agent run`. FR-017 must not reposition `cic agent orchestrate` as the default.

4. Audit inventory (derive-only proof)

4.1 Ownership matrix (normative)

Concern	Owner FR	FR-017 role
Workflow traces / checkpoints / retries	FR-008	None — do not re-instrument
Unsupported-claim / truth	FR-014	None — do not duplicate
BOPA metrics / agent corpus	FR-015 M4 (<code>agent.observability</code>)	Reuse child AgentRun metrics
Parent/child audit, handoffs, loop controls	FR-016	Source for derived orchestration metrics
Orchestration metrics API + eval packaging	FR-017 (proposed)	Derive + aggregate + harness only
Dashboards / SaaS observability	—	Out of scope
Browser / Playwright journeys	Deferred acquisition	Out of scope

4.2 OrchestrationRun fields → proposed metrics

Existing field / structure	Derivable metric / reconstructability signal
<code>orchestration_run_id</code> , <code>goal</code>	Identity; owner goal kind / flags
<code>status</code> , <code>stop_reason</code> , <code>owner_action_required</code>	Outcome; owner next step
<code>step_count</code> , <code>max_steps</code>	Step utilisation
<code>max_visits_per_specialist</code> , <code>specialist_visits</code>	Visit counts; visit-limit proximity
<code>handoff_ids</code> + loaded <code>Handoff</code> records	Handoff count; allow/deny; lifecycle
<code>child_agent_run_ids</code>	Child BOPA refs (join to FR-015 metrics)
<code>last_brief_id</code>	OBS output ref
<code>last_observation</code>	Observed state class, package/truth/pipeline, hash
<code>events[]</code> (<code>OrchestrationAuditEvent</code>)	Timeline; selection; policy; specialist start/stop
<code>created_at</code> , <code>updated_at</code>	Elapsed time
<code>checkpoint_ref</code>	Recovery pointer (not SoT)
<code>owner_approvals_present</code> , <code>provider_available</code>	Gate / availability flags

4.3 Handoff fields → reconstructability

Field	Reconstructability use
<code>source</code> (always <code>supervisor</code>)	Proves DOS-sourced delegation
<code>target_specialist</code> , <code>requested_goal_kind</code>	Authority target
<code>reason</code>	Selection rationale
<code>policy_decision</code> , <code>policy_deny_reason</code>	DelegationPolicy result
<code>acceptance</code> , <code>acceptance_reason</code>	Lifecycle
<code>observed_state_hash</code> , <code>idempotency_key</code>	Idempotency / no-dupe proof
<code>expected_output_kind</code>	Contracted output
<code>child_agent_run_id</code> / <code>child_brief_id</code>	Specialist result linkage
<code>created_at</code> , <code>resolved_at</code>	Handoff latency (derived)

4.4 Child AgentRun (FR-015 — reuse, do not fork)

When `child_agent_run_ids` is non-empty, call existing `career_intelligence.agent.observability.extract_run_metrics` — tokens / cost / provider / action allow-deny already defined. FR-017 must **not** redefine BOPA metrics.

4.5 Derive-only gate (M1 entry criterion)

Pass condition: Every M1 metric is expressible as a pure function of stored `OrchestrationRun` + `Handoff` (+ optional child `AgentRun`) with **zero** new runtime event kinds and **zero** DOS/BOPA/OBS behaviour changes.

Fail condition: Any metric that requires new instrumentation, new SoT, or policy change → drop the metric or NO-GO that slice.

M0 result: Gate **passes** for the narrow metric set in §6.

5. Strengthened audit-reconstructability criteria

These are **first-class FR-017 acceptance criteria** (stronger than “show elapsed time”). A frozen FR-017 must prove that from package artefacts + audits alone, without reading implementation source, an engineer can answer:

#	Reconstructability question	Evidence source
R1	What owner goal started the run?	OrchestrationGoal
R2	What authoritative state was observed (decision, package, truth, pipeline, hash)?	last_observation / events
R3	Which specialists were candidates vs selected, and why?	events + handoff reason
R4	Did DelegationPolicy allow or deny, and why?	handoff policy_*
R5	What ToolPolicy boundary applied to the selected specialist?	specialist registry + presentation/authority (existing); metrics cite specialist id
R6	What was the handoff lifecycle?	acceptance trail
R7	What child AgentRun or OperationalBrief resulted?	child ids
R8	Why did orchestration stop?	stop_reason
R9	What must the owner do next?	owner_action_required
R10	Were global step / visit limits approached or hit?	step/visit fields
R11	Can parent → handoff → child be walked without gaps?	handoff_ids + child refs + events
R12	On resume, was SoT re-inspected (hash change) without duplicate specialist work?	observation hashes + idempotency keys + child completed ops

Definition of reconstructability for FR-017: R1–R12 are demonstrable via derived metrics + existing `show / history` presentation for at least the FR-016 corpus cases A–T (or an explicit subset documented in M1), offline, with no live LLM required.

Missing R-signals that cannot be derived → either accept as documented limitation or extend **presentation only** — never by inventing silent domain side effects.

6. Narrow scope contract (in / out)

6.1 In scope for M1+ (if owner accepts GO)

1. `multi_agent.observability` — pure derive of `orchestrationRunMetrics` / corpus aggregates (mirror FR-015 shape, orchestration vocabulary).
2. Join child AgentRun metrics via existing FR-015 helpers when children exist.
3. Package FR-016 corpus + fault-injection cases as an explicit eval API surface (thin wrappers / tests — behaviour unchanged).

4. Optional read-only CLI: `cic agent orchestrate metrics <orr_...>` (and/or list aggregates) — thin; no dashboard.
5. Documentation: ADR (if needed), M1–M4 eval trail, Academy package after freeze.
6. Reconstructability suite proving R1–R12.

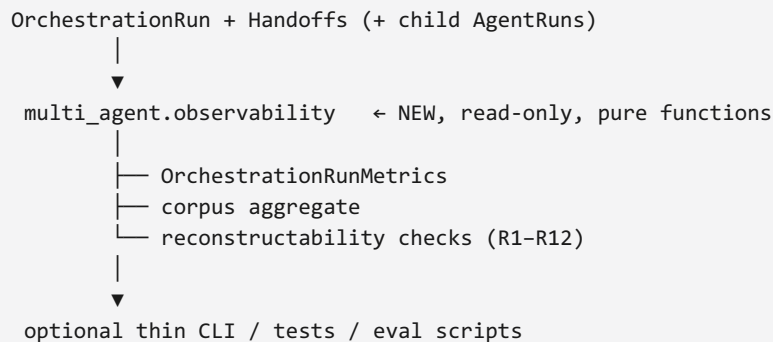
6.2 Explicitly out of scope

- Dashboards, chart UIs, SaaS backends (LangSmith-like)
- Framework adoption
- DOS / BOPA / OBS redesign; ToolPolicy / DelegationPolicy changes
- New orchestration audit event kinds (unless M1 proves a genuine gap against R1–R12)
- FR-008 re-instrumentation
- FR-014 claim engine duplication
- Browser / Playwright journey evidence
- Live LLM supervisor evaluation as a core exit criterion
- Making orchestration the daily prep default
- Job Discovery / Horizon 1B features
- Blocking Horizon 1B on FR-017 completion

6.3 Laundry-list disposition (functional spec text)

Spec phrase	Disposition
traces / checkpoints / retries	Already FR-008 — out
golden workflow tests	Already FR-008 — out
loop prevention	Already FR-016 — out (metrics may <i>report</i> limits)
unsupported-claim checks	Already FR-014 — out
token/latency/cost	Derive from child AgentRun <code>ProviderMetadata</code> when present; null offline OK
fault injection / orchestration testing	In as harness packaging of existing patterns
deterministic replay	In as offline static-observation re-run (corpus style) — not live SoT time-travel
browser journey evidence	Out
approval interrupts	Already owner <code>--approve</code> — metrics may flag <code>owner_approvals_present</code>

7. Recommended architecture (M1 sketch — not implementation)



Invariant: No write path into Opportunity, package, truth, pipeline, or submission SoTs. No DOS behavioural change.

8. Risks

Risk	Mitigation
Observability theatre	Out-of-scope list; M1 ADR/notes forbid dashboards
Scope creep into FR-008/014/015	Ownership matrix §4.1
FR-016 redesign under metrics pressure	Derive-only gate §4.5
Blocking 1B	Normative: FR-017 must not block Horizon 1B (§9)
Overclaiming commercial value	Learning/substrate posture explicit
Reconstructability theatre (checklists without evidence)	R1–R12 must be demonstrated on corpus

9. Horizon 1B coupling (normative spike decision)

Decision: Horizon 1B (FR-018+) is **not** blocked on FR-017.

Rationale:

- Horizon 1 sequencing principle is **job acquisition first**.
- The discover → assess → prepare → review → submit → track loop is already delivered by FR-008–FR-015.
- FR-016 is an optional learning proof, not the daily prep path.
- FR-017 (narrow) improves **evaluation substrate**, not application throughput.

Usable Horizon 1A for unblocking 1B means: application loop works under owner approval and truth gates — **not** “every learning FR including eval is frozen.”

Roadmap / functional specification must be amended accordingly (this M0). Owner may still choose to finish FR-017 before starting 1B for learning continuity; that is preference, not an engineering gate.

10. Complexity assessment

Slice	Effort	Notes
Derive observability module + tests	Low–medium	Mirror FR-015
Reconstructability assertions on corpus	Low–medium	Strengthen quality bar
Thin metrics CLI	Low	Optional in M2/M3
Full laundry list / dashboard	High	Rejected

11. Learning value

FR-017 (narrow) teaches:

- evaluation as **derived views over audits**
 - reconstructability as an acceptance criterion, not a slide
 - continuity with FR-015 observability and FR-016 authority lessons
 - why dashboards are not synonymous with observability
 - honest low commercial value vs high interview transferability
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12. Academy workflow integration

No process redesign. After FR-017 freeze:

`docs/masterclass/FR017/` via `scripts/build_masterclass_package.py`
SoT remains `docs/eval/` + ADR. Package is educational packaging only.

13. Go / no-go

Option	Decision
Full functional-spec laundry list	NO-GO
Dashboard / framework observability	NO-GO
FR-017 blocks Horizon 1B	NO-GO (coupling rejected)
Narrow derive-only eval + R1–R12 reconstructability	GO to M1

Binding posture: learning / substrate; daily prep unchanged; derive-only; reconstructability-first; 1B not gated.

14. Proposed milestones (after owner accepts M0 GO)

Milestone	Intent
M0	This spike — complete
M1	Contracts: <code>OrchestrationRunMetrics</code> , derive API, R1–R12 check helpers, unit tests; optional ADR-009 if decisions need freezing
M2	Wire corpus aggregation + reconstructability suite on FR-016 cases; no DOS changes
M3	Thin owner CLI for metrics (optional if M2 evidence shows CLI unnecessary)
M4	Evaluation, docs freeze, Academy package

Do not begin M1 until owner explicitly accepts this M0 GO.

15. Definition of Done (M0)

Criterion	Status
Inventory of existing audits	Done (§4)
Ownership matrix	Done
Derive-only gate defined and passed for narrow set	Done
Reconstructability criteria R1–R12	Done (§5)
In/out scope contract	Done (§6)
Horizon 1B decoupling	Done (§9)
Go/no-go	GO to M1 (narrow)
No production code	Confirmed

16. Owner next step

Done: Owner accepted M0 GO (narrow scope) and unlocked M1.

Next: Accept M1 contracts freeze to unlock M2 (corpus + reconstructability suite) — **or** defer FR-017 and proceed to Horizon 1B (allowed by §9).

Do not implement M2 without explicit owner request.